① model: $L^{\lambda\lambda'}_{\varepsilon p} = \sqrt{\gamma^{\lambda\lambda'}_{\varepsilon p}}\; d^\dagger_{\varepsilon\lambda}\, d_{p\lambda'}$ $\qquad \mu,\nu,\lambda,\lambda' = \{\alpha,\beta\}$ indeks $\qquad d_i = \{\alpha_i, \beta_i\}$ qubits

$\Gamma^{\lambda\lambda'}_{kp}$:

$$\Gamma^{\mu\nu}_{kp} = \varkappa\, e^{-\left(\frac{E_{\mu k}-E_{\nu p}}{\omega_c}\right)^2} \cdot \frac{1}{1+e^{\frac{E_{\mu k}-E_{\nu p}}{T}}} \to \Delta E$$

$\mu(k) \leftarrow \nu(p)$ $\qquad$ skalar.

$$\frac{\Gamma^{\mu\nu}_{kp}}{\Gamma^{\nu\mu}_{pk}} = \frac{\cancel{\varkappa e^{\dots}}}{\cancel{\varkappa e^{\dots}}} \cdot \frac{\frac{1}{1+e^{(E_{\mu k}-E_{\nu p})/T}}}{\frac{1}{1+e^{(E_{\nu p}-E_{\mu k})/T}}} = \frac{\frac{1}{\cancel{1+e^x}}}{\frac{1}{\cancel{(1+e^{-x})e^x}}\,e^x} = e^{-x} = e^{-(E_{\mu k}-E_{\nu p})/T}$$

$$\to \boxed{\Gamma^{\mu\nu}_{kp} = \Gamma^{\nu\mu}_{pk} \cdot e^{-(E_{\mu k}-E_{\nu p})/T}} \quad \Bigg\} \frac{\mu(k) \leftarrow \nu(p)}{\mu(k) \rightarrow \nu(p)}$$

② model

$$\Gamma^{\mu\nu}_{kp} = \varkappa^{\mu\nu}\, e^{-\left(\frac{\Delta E}{\omega_c}\right)^2} f\left(\frac{E_{\mu k}-\mu}{T}\right) f\left(-\frac{E_{\nu p}-\mu}{T}\right)$$

$$1 - f(E) = f(-E)$$

$$\frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x} = 1 - \frac{1}{1+e^x}$$

DB: $\varkappa^{\mu\nu} = \varkappa^{\nu\mu}$

$$\frac{\Gamma^{\mu\nu}_{kp}}{\Gamma^{\nu\mu}_{pk}} = \frac{f(x)\,f(-y)}{f(-x)\,f(y)} = \frac{\frac{1}{1+e^x}\cdot\frac{1}{1+e^{-y}}\cdot\frac{e^y}{e^y}}{\frac{1}{1+e^{-x}}\cdot\frac{1}{1+e^y}\cdot\frac{e^x}{e^x}} = e^{-(x-y)} = e^{-(E_{\mu k}-\mu-E_{\nu p}+\mu)/T} =$$

$$= e^{-\Delta E/T} \checkmark$$

$$x = \frac{E_{\mu k}-\mu}{T} \qquad y = \frac{E_{\nu p}-\mu}{T}$$

~ Thermalization at Low T via Weakly dampened Multi site Baths:

- goal: $\{L_j^k\}$, ki peljejo kopel v GE, kjer fiksni T_B in μ_B:
  $\hookrightarrow \rho_B = \frac{1}{Z} e^{-(H_B - \mu_B N_B)/T_B}$ $\quad \underbrace{N_B}_{\#\text{ delcev}} \to \mathcal{L}_B \rho_B \overset{!}{=} 0$.

- impose DB za spekter H_B.

→ noninteracting / diagonalizabilni sistemi:

$$H_B \to J^B = V^\dagger \varepsilon^B V, \quad \varepsilon^B = \mathrm{diag}(\varepsilon^B_1, \dots, \varepsilon^B_n), \quad H_B = \vec{c}^{\,\dagger} \vec{c}\, J_B =$$

$$= \sum_k V^\dagger c_k^\dagger c_k \varepsilon^B_k V$$

$\{c_k, c_e^\dagger\} = \delta_{ke}$

$$\to \rho_B = \prod_{k=1}^{M} \left(f_k c_k^\dagger c_k + (1-f_k)\, c_k c_k^\dagger\right) \qquad f_k = \frac{1}{1+e^{(\varepsilon_k-\mu)/T}}$$

$\to L_k^{in} = \sqrt{\gamma f_k}\, c_k^+ \qquad L_k^{out} = \sqrt{\gamma(1-f_k)}\, c_k \Big\}$ A5 + A6 : Taki operatorji zagotovijo:

$\mathcal{L}_B \rho_B = 0$!

(3a) bozonski bath:

$H_B = \sum_q \omega_q b_q^+ b_q \,, \quad \rho_B \simeq e^{-\beta H_B} \,, \quad \bar{n}_B = \frac{1}{e^{\omega_q/T} - 1}$

$L_q^- = \sqrt{\gamma_q(\bar{n}_q+1)}\, b_q \,, \quad L_p^+ = \sqrt{\gamma_p \bar{n}_p}\, b_p^+ \to L_{qp}^{++,+-,-+,--} = L_q^{(\pm)} L_p^{(\pm)}$

$\mathcal{D}[\circ] = \sum_j L_j \rho L_j^+ - \frac{1}{2}\{L_j^+ L_j, \rho\}$

(1) ~~qu~~ el.

$H_{SB} = \sum_{ij} g_{ijq}\, c_i^+ c_j (b_q + b_q^+)$